

## ■ Example: RSA with small numbers

ALICE

Message  $x = 4$

BOB

1. Choose  $p = 3$  and  $q = 11$
2. Compute  $n = p * q = 33$
3.  $\Phi(n) = (3-1) * (11-1) = 20$
4. Choose  $e = 3$
5.  $d \equiv e^{-1} \equiv 7 \text{ mod } 20$

$K_{\text{pub}} = (33, 3)$



$$y = x^e \equiv 4^3 \equiv 31 \text{ mod } 33$$

$y = 31$



$$y^d = 31^7 \equiv 4 = x \text{ mod } 33$$